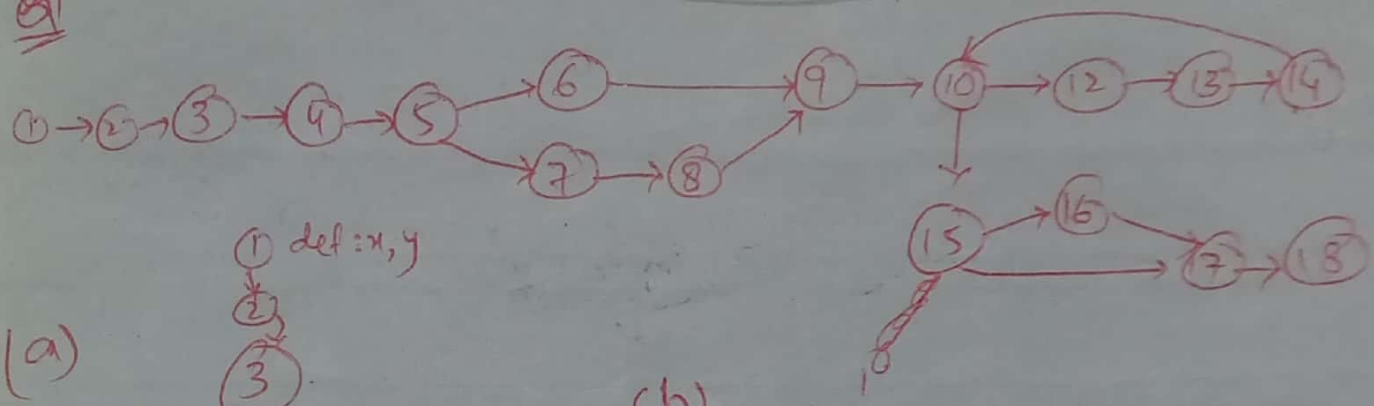
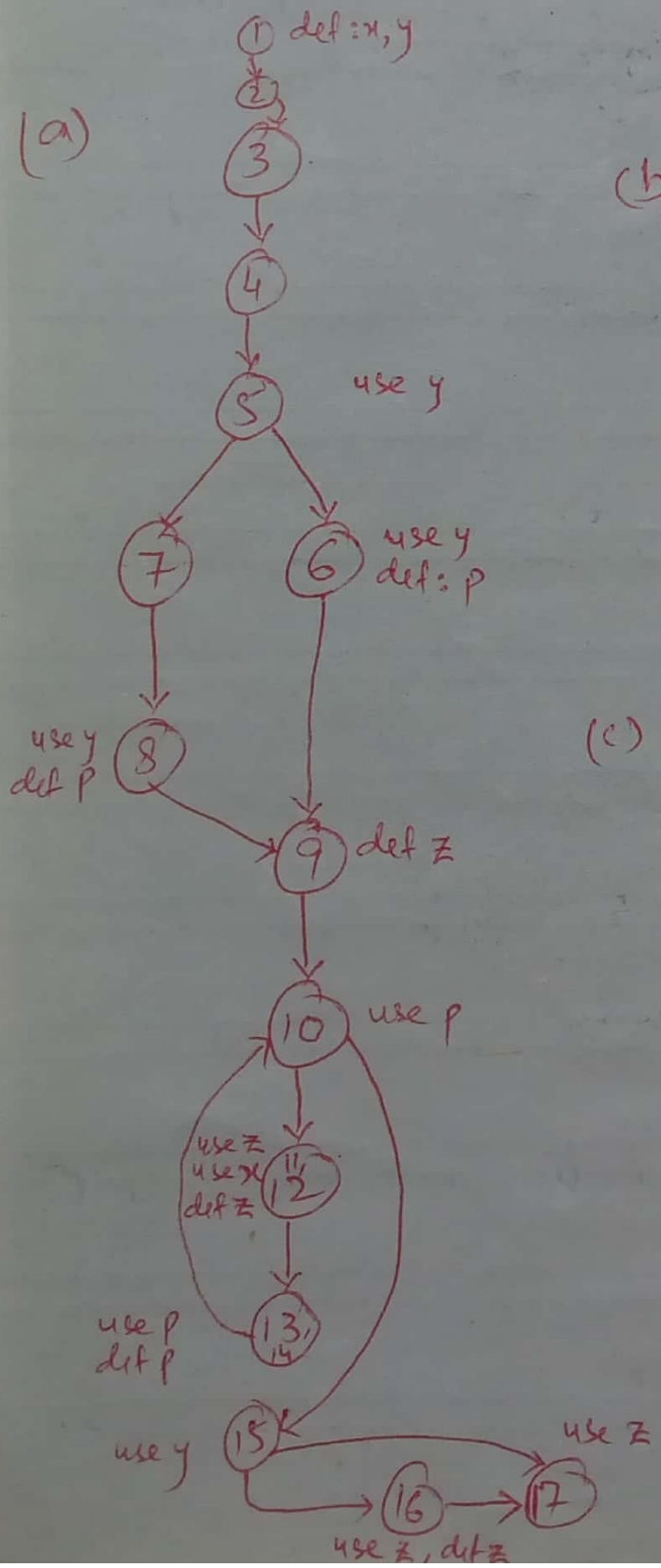
Q1



(a)



(b)

P1: 1,2,3,4,5

P2: 1,2,3,4,5,6

P3: 1,2,3,4,5,7,8

P4: 1,2,3,4,5,6,9,10,15

P5: 1—6,9,10,12,13,10,15

P6: 1,2,3,4,5,6,9,10,12

P7: 1,2—6,9,10,12,13,10,12

P8: 1,2,3,4,5,7,8,9,10,12

P9: P8' (with loop)

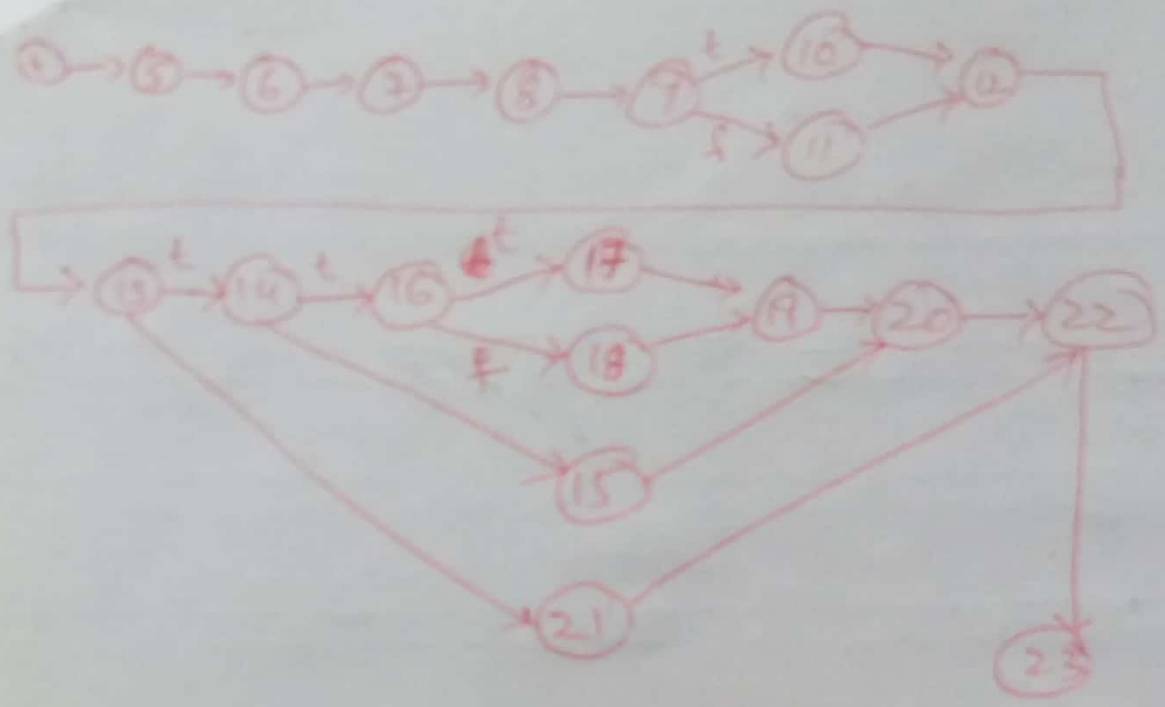
P10: 1-5,7,8,9,10,15

P11: 1-5,7,8,9,10,12,13,15

(c) All paths are dc paths.

Software Testing Mid 1 Solution

Spring 2020



(b)

$E - N + 2 = 23 - 20 + 2 = 5$

$P + 1 = 4 + 1 = 5$

Regions $= 5$

(c) ① 4–9, 10, 12, 13, 14, 16, 17, 19, 20, 22, 23 (infeasible)

② 4–9, 11, 12, 13, 14, 16, 17, 19, 20, 22, 23 (infeasible)

③ 4–9, 10, 12, 13, 21, 22, 23 (infeasible)

④ 4–9, 10, 12, 13, 16, 15, 20, 22, 23 (equilateral)

⑤ 4–9, 10, 12, 13, 14, 16, 18, 19, 20, 22, 23 (feasible)

(d)

	a	b	c
P1	4	4	4
P2	1	2	3
P3	—	—	—
P4	3	4	5
P5	3	3	4

P1: 4–9, 10, 12, 13, 14, 15, 20, 22, 23 (Equilateral)

P2: 4–9, 10, 12, 13, 21, 22, 23 (Not a Δ)

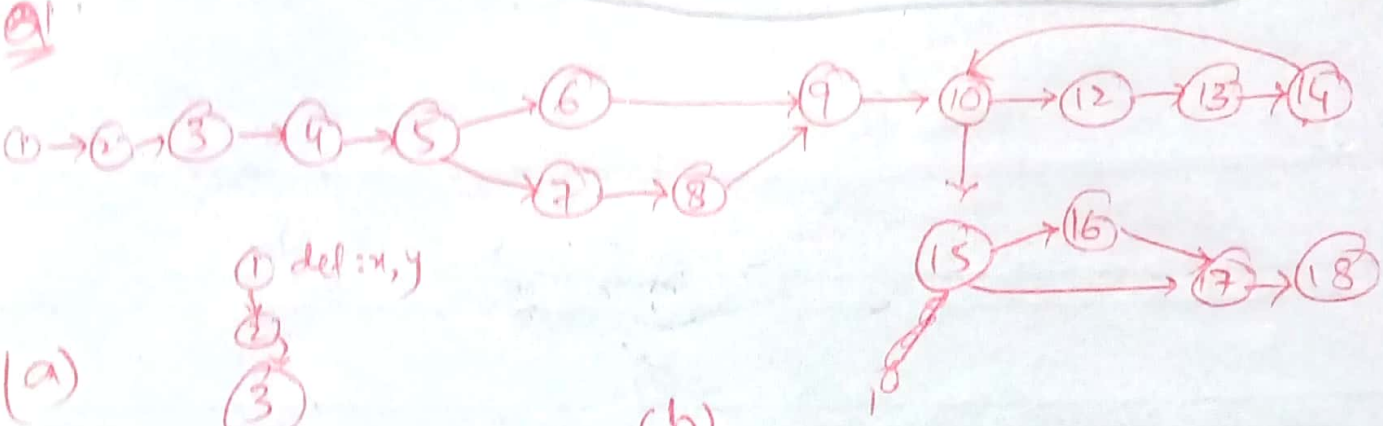
P3: 4–9, 10, 12, 13, 21, 22, 23 (infeasible)

P4: 4–9, 10, 12, 13, 14, 16, 17, 19, 20, 22, 23 (feasible)

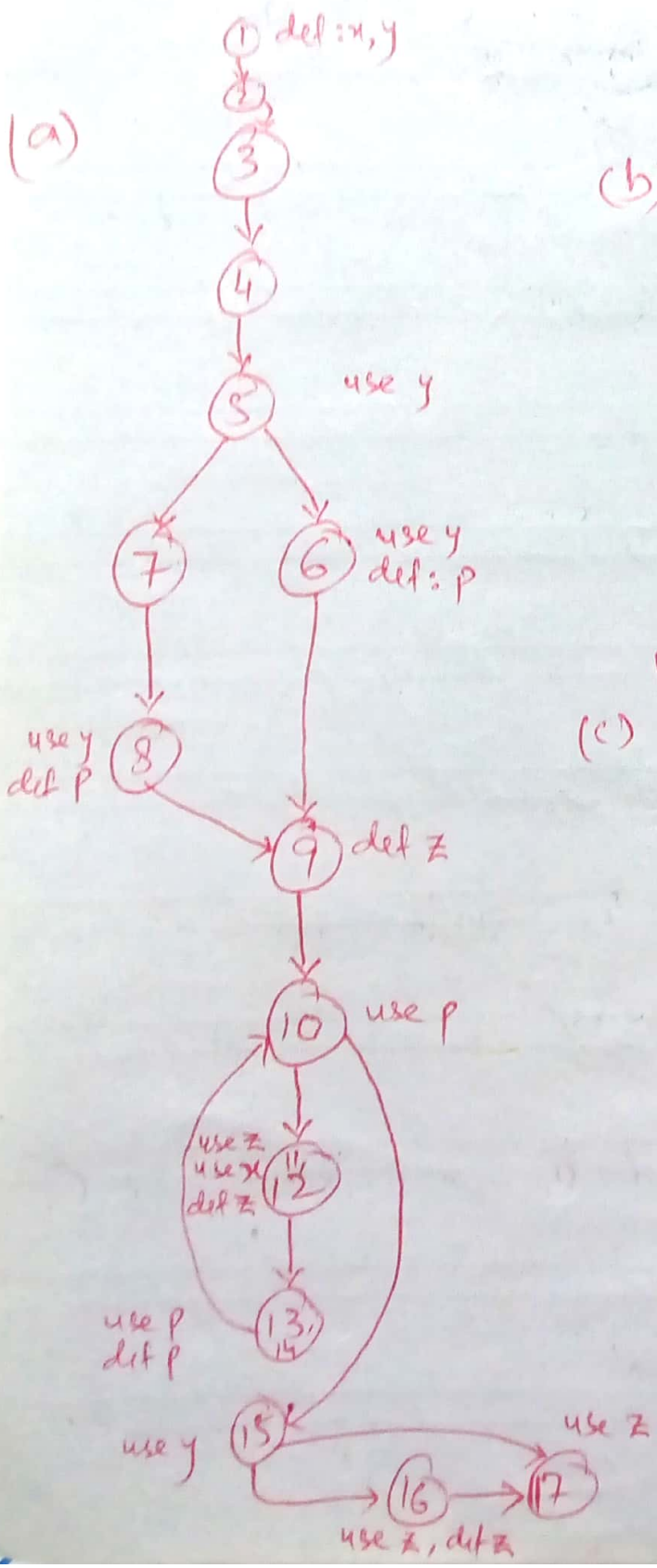
P5: 4–9, 10, 12, 13, 14, 16, 18, 19, 20, 22, 23 (feasible)

Scanned with CamScanner

Q1



(a)



(b)

P1: 1,2,3,4,5

P2: 1,2,3,4,5,6

P3: 1,2,3,4,5,7,8

P4: 1,2,3,4,5,6,9,10,15

P5: 1—6,9,10,12,13,10,15

P6: 1,2,3,4,5,6,9,10,12

P7: 1,2—6,9,10,12,13,10,12

P8: 1,2,3,4,5,7,8,9,10,12

P9: P8' (with loop)

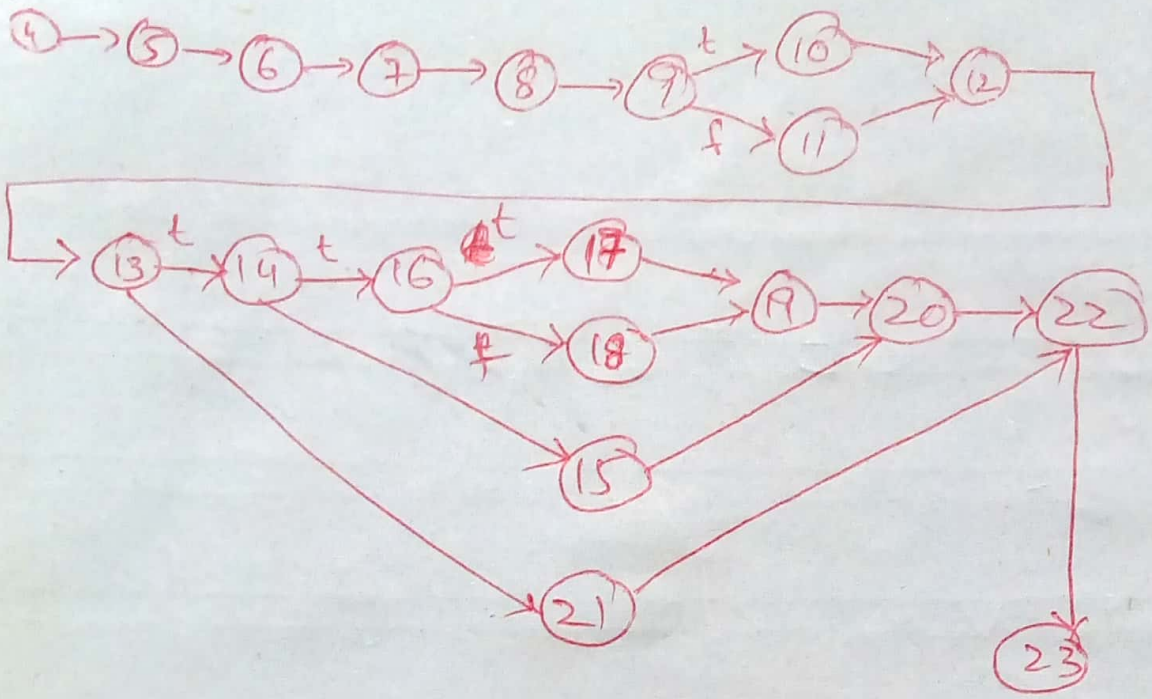
P10: 1-5,7,8,9,10,15

P11: 1-5,7,8,9,10,12,13,15

(c) All paths are de paths.

Scanned with CamScanner

Software Testing
Mid 1
Solution
Spring 2020



(b)

$E - N + 2 = 23 - 20 + 2 = 5$

$P + 1 = 4 + 1 = 5$

Regions $= 5$

(c) ① 4–9, 10, 12, 13, 14, 16, 17, 19, 20, 22, 23 (infeasible)

② 4–9, 11, 12, 13, 14, 16, 17, 19, 20, 22, 23 (infeasible)

③ 4–9, 10, 12, 13, 21, 22, 23 (infeasible)

④ 4–9, 10, 12, 13, 14, 15, 20, 22, 23 (equilateral)

⑤ 4–9, 10, 12, 13, 14, 16, 18, 19, 20, 22, 23 (Isosceles)

(d)

	a	b	c
P1	4	4	4
P2	1	2	3
P3	–	–	–
P4	3	4	5
P5	3	3	4

P1: 4–9, 10, 12, 13, 14, 15, 20, 22, 23 (Equilateral)

P2: 4–9, 11, 12, 13, 21, 22, 23 (Not a Δ)

P3: 4–9, 10, 12, 13, 21, 22, 23 (infeasible)

P4: 4–9, 10, 12, 13, 14, 16, 17, 19, 20, 22, 23 (Scalene)

P5: 4–9, 10, 12, 13, 14, 16, 18, 19, 20, 22, 23 (Isosceles)

Scanned with CamScanner